The Cartagena Protocol on Biosafety

Safeguarding Biodiversity in the Age of Biotechnology

An international framework protecting our planet's biological diversity from potential risks of genetic modification while enabling safe biotechnology innovation.

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Origins of the Protocol: A Global Response to Biotechnology Risks

1

1996

Negotiations launched under the Convention on Biological Diversity to address growing concerns about biotechnology's environmental impacts

2

January 2000

Protocol adopted in Montreal, Canada, named after Cartagena, Colombia, where breakthrough discussions occurred

3

September 2003

Entered into force after achieving 50 ratifications, establishing binding international biosafety standards

4

Today

173 Parties committed, including 170 UN member states and the European Union

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What is the Cartagena Protocol?

Core Purpose

An international treaty supplementing the Convention on Biological Diversity, establishing a comprehensive regulatory framework for biosafety.

The Protocol regulates transboundary movement, handling, and use of living modified organisms to ensure biotechnology develops safely.

Protects Biodiversity

Safeguards ecosystems from potential adverse effects of genetically modified organisms

Human Health Focus

Ensures LMOs don't pose risks to human populations and food security

Precautionary Approach

Applies Rio Declaration Principle 15: act with caution when scientific uncertainty exists

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Defining Living Modified Organisms (LMOs)

What Are LMOs?

Living organisms with novel genetic combinations created through modern biotechnology techniques, resulting in characteristics that don't occur naturally.

Engineered Crops

Corn, soybeans, cotton, and rice modified for pest resistance or enhanced nutrition

Microorganisms

Bacteria and fungi engineered for industrial or environmental applications

Modified Animals

Fish and livestock with enhanced growth or disease resistance traits

Important exclusion: The Protocol covers LMOs intended for environmental release but excludes pharmaceuticals for humans and processed products without living genetic material.

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Key Mechanisms: Advance Informed Agreement (AIA)

01

Export Notification

Exporting country notifies importing country before first LMO shipment intended for environmental release

02

Information Sharing

Scientific, technical, and risk assessment data provided to support informed decision-making

03

Import Decision

Importing country reviews information and makes science-based decision to approve, reject, or request modifications

04

Consent Required

Shipment proceeds only after explicit written consent from importing country authorities

> "The AIA procedure empowers nations—especially developing countries—to balance potential economic benefits of biotechnology against biosafety concerns and local environmental conditions."

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The Biosafety Clearing-House (BCH)

A Global Information Hub

An internet-based platform serving as the Protocol's central mechanism for information exchange and transparency.

- Facilitates sharing of scientific, technical, environmental, and legal information on LMOs
- Provides access to national decisions, risk assessments, and regulatory frameworks
- Supports capacity building among Parties through training resources
- Enables real-time tracking of LMO approvals and regulations worldwide

173

Countries Connected

Parties accessing and contributing biosafety information

24/7

Access Available

Continuous information availability for decision-makers

1000s

Documents Shared

Risk assessments, laws, and scientific studies

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Trade and Labeling Provisions

Documentation Requirements

Bulk shipments of LMO commodities (corn, soybeans, canola) must carry clear documentation stating they **"may contain"** living modified organisms.

Intended Use Declaration

Shipping documents must specify LMOs are **"not intended for intentional introduction into the environment"** but rather for food, feed, or processing.

Balancing Trade and Safety

- Protocol respects existing WTO rules and international trade obligations
- Does not mandate physical segregation of LMO from non-LMO shipments
- No requirement for consumer product labeling at retail level
- Focuses on transboundary movement documentation rather than downstream tracking

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Impact and Global Reach

173 Parties

United in their commitment to biosafety and biodiversity protection across six continents

Capacity Building

Technical assistance and training programs help developing countries establish robust biosafety frameworks and regulatory infrastructure

Risk Assessment

Standardized methodologies enable countries to conduct science-based evaluations of LMO safety

Precautionary Governance

Model framework balancing innovation with environmental protection through transparent, evidence-based decision-making

Safe Innovation

Facilitates responsible biotechnology use while preventing environmental harm and protecting ecosystems

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Challenges and Future Directions

Enhanced Identification

Ongoing negotiations on precise identification and documentation of LMO shipments to improve traceability

Standardized Assessment

Developing harmonized risk assessment methodologies applicable across diverse ecosystems and contexts

Innovation Balance

Addressing rapid biotech advances while maintaining environmental protection amid gene editing and synthetic biology

Strengthening the Framework

Compliance

Enhancing monitoring mechanisms and enforcement tools

Public Awareness

Expanding education and stakeholder engagement globally

Technical Support

Increasing resources for developing nations' biosafety programs

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Protecting Our Planet's Biodiversity Together

Pioneering Framework

The Cartagena Protocol stands as the world's first comprehensive global agreement ensuring biotechnology develops safely and responsibly

Empowered Nations

Countries worldwide are equipped to protect biodiversity and human health through science-based, transparent decision-making processes

Shared Responsibility

Continued international cooperation and commitment remain vital to address emerging biosafety challenges in our rapidly evolving world

Together, we harness biotechnology's benefits while safeguarding our shared environment

The Protocol's success depends on collective action—governments, scientists, industry, and civil society working in partnership to ensure a sustainable, biodiverse future for generations to come.

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